



Bio-inspired design of antifouling polymeric coatings with natural extracts: key evidence for resistance to fouling adhesion

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Abstract

Marine environmentally friendly antifouling materials are emerging as a viable and promising alternative to the conventional toxic antifouling agents. Within this field, the exploration of natural antifouling substances has become a significant research focus, representing an auspicious avenue for innovation in eco-friendly technologies. In this study, we delve into the development of eco-friendly antifouling coatings through a novel chemical modification process. By incorporating natural antifouling agents onto an acrylic acid substrate through a grafting process, we have successfully synthesized three distinct varieties of natural antifouling coatings: isobornyl acrylate polymer (IBAP), acrylate indole polymer (AIP), and indole isobornyl co-modified acrylate polymer (IAA-IBOMA). Through meticulous surface characterization, structural analysis, and a comprehensive suite of antifouling performance tests, our findings indicate that these coatings exhibit superior antifouling properties. Notably, the IAA-IBOMA coating demonstrated exceptional anti-adhesion effects. The specific inhibition rates against *E. coli*, *S. aureus*, and *Pseudoalteromonas aeruginosa* were impressive, achieving 93.5%, 92.8%, and 95.7%, respectively. Moreover, the anti-mussel selective adhesion inhibition rate was found to be 93.3%. Furthermore, environmental toxicity assessments have validated the eco-friendly and stable nature of the IAA-IBOMA coating. These results underscore the potential of these natural product-based coatings as sustainable solutions for the marine industry. This work offers valuable insights and holds significant implications for guiding the future development of environmentally friendly antifouling coatings, steering the industry towards a more sustainable and eco-conscious direction.

Keywords Antifouling coatings · Natural extracts · Isoborneol · Indole anti-adhesion effects

1 Introduction

Marine biofouling [1, 2] can lead to huge economic losses [3, 4] and serious damage to the ecological environment [5, 6]. To solve the marine biofouling problem, the most widely used and effective strategy is antifouling coating technology [7, 8]. However, traditional antifouling coatings contain organotin compounds (e.g., tributyltin oxide and tributyltin furan), copper compounds, etc., which have persistent toxicity and pose a serious threat to marine organisms and marine environmental protection. Therefore, there is an urgent need to develop environmentally friendly antifouling coatings [9–11]. In recent years, fluorescent antifouling technology has attracted widespread attention because of its environmental protection and high efficiency. The technology utilizes the special properties of fluorescent substances to achieve antifouling effects by interfering with or inhibiting the attachment of marine organisms on the material surface. This antifouling method is not only environmentally friendly

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but also has a wide range of application prospects, especially in the field of marine engineering, ship coatings, etc. [12].

Furthermore, in the quest for sustainable solutions, natural products derived from marine microorganisms, algae, aquatic plants, marine invertebrates, as well as terrestrial sources have risen to prominence as promising antifouling agents [13]. These substances offer a viable and environmentally conscious alternative to traditional antifouling methods, highlighting their significant role in shaping the future of marine preservation and material protection [14]. These compounds include fatty acids, lactones, terpenes, steroids, benzenes, phenyl ethers, polyketides, alkaloids, nucleosides, and peptides [15, 16]. They not only have strong antifouling properties but are environmentally friendly. As an example, 6-bromoindole-3-carboxaldehyde and 2,5,6-tribromo-1-methylrutacine [17], two natural indole derivatives extracted from sea squirts and moss worms, have shown excellent antifouling ability for barnacles and mussels [18, 19]. Kangdi Hu et al. [20] synthesized indoles containing carbon–carbon double bonds from 5-Cl/NO₂-indole, which showed antimicrobial activity against typical Gram-negative (*Escherichia coli*) and Gram-positive (*Staphylococcus aureus*) bacteria with an antimicrobial rate of 92.79% and 99.71% and 90.35% inhibition against algae, respectively. The above studies showed that the indole derivatives not only had high antifouling activity but also were environmentally friendly.

Currently, the ice tablet base has attracted much attention in the preparation of marine coatings. Ice tablet (borneol, C₁₀H₁₈O), also known as lobelia [21], is chemically composed of borneol, and the molecular structure contains a six-membered carbon skeleton with a number of methyl groups attached to it [22]. The cage structure of bingshi contains many chiral carbon atoms, so bingshi itself has many isomers. Bingshi has been used as a medicine in the Materia Medica Compendium for dispersing fire, killing worms, improving eyesight, and clearing heat [23]. In addition, studies have also found that ice chips have excellent antibacterial effects [24, 25]. Low concentrations of ice chips can inhibit bacterial growth and act as a bacteriostat [11, 26, 27], while high concentrations can kill bacteria and act as a sterilizing agent [28]. For example, Susu Zhao et al. [29] presented that the synergistic effect of natural ice flakes with the dodecafluoroheptyl group improved the electrostatic antifouling performance of silicone coatings, and the antimicrobial rate against marine bacteria (*Pseudomonas* sp.) reached 95.6%, which natural camphor was added as an antifouling agent. The authors investigated the relationship between the content of IBOMA and the antifouling performance and found that the diatom adhesion inhibition performance of the coating was significantly improved as the content of IBOMA was increased from 16.7 to 50%; the excellent anti-adhesion performance of IBOMA was demonstrated by antimicrobial, antialgae,

and real-sea pegboard. This research result provides new ideas and methods for the development of efficient and environmentally friendly marine antifouling coatings [30].

In this study, free radical synthesis was utilized to graft indole and icosahedral monomers on acrylic acid and indole co-modified polymers on acrylic icosahedral. The coatings were isobornyl acrylate polymer (IBAP), acrylate indole polymer (AIP), and indole isobornyl co-modified acrylate polymer (IAA-IBOMA), respectively. The polymer structures were investigated by FT-IR and ¹H NMR, which proved that the synthesis of the natural extract polymers was successful. The smoothness and flatness of the coating surfaces were demonstrated by SEM, AFM, and CA tests, and these coatings were proved to have good antifouling performance, i.e., environmentally friendly, by antimicrobial, algacidal, and mussel selective adhesion resistance experiments, and by environmental toxicological experiments. This work centers on environmentally friendly marine antifouling coatings to carry out experiments to provide favorable evidence for marine biofouling and to achieve high efficiency, low toxicity, and green antifouling goals, which is of great significance for marine ecological environment protection.

2 Experimental section

2.1 Experimental reagents and equipment

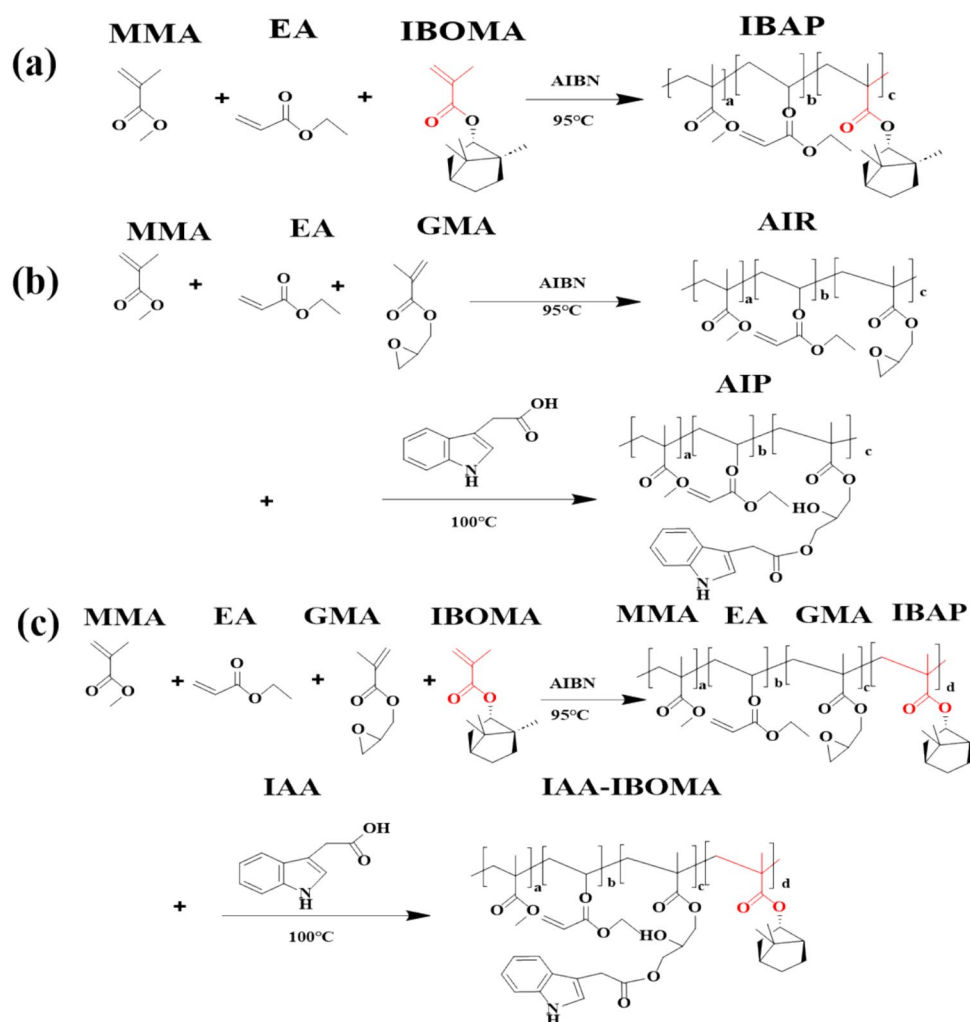
The analytical purity of isobornyl methacrylate (IBOMA), indoleacetic acid (IAA), benzoyl peroxide (BPO), glycerol methacrylate (GMA), azobisisobutyronitrile (AIBN), methyl methacrylate (MMA), and ethyl acrylate (EA) were purchased from McLean Co. The analytical purity of sodium chloride, peptone, and agar was purchased from Aladdin Co. Deionized water was made by the laboratory. All other reagents were used directly without treatment. FT-IR spectrometer Spectrum100 PerkinElmer, fluorescence microscope OLYMPUS-BX51 Japan, UV spectrophotometer Shanghai Jingke, light incubator GXZ-400B Ningbo Southeast Instrument Co. Ltd., and the electric thermostat were from Shanghai Xu Ri Instrument Co.

2.2 Synthesis of natural extract-based antifouling coatings

2.2.1 Synthesis of isobornyl acrylate polymer (IBAP)

As shown in Scheme 1a, first, 28 g of xylene and 12 g of propylene glycol methyl ether were mixed with acrylic acid as the substrate and heated to 95 °C in a four-necked flask. Next, 47 g of methyl methacrylate (MMA), 23 g of ethyl acrylate (EA), 30 g of isobornyl methacrylate (IBOMA), 1 drop of a chain transfer agent (MSD), and 1.5 g of azobisisobutyronitrile (AIBN) were mixed with 28 g of xylene and 12 g of

Scheme 1 Synthesis of IBAP coatings (a), AIP coatings (b), and IAA-IBOMA coatings (c)



propylene glycol methyl ether and stirred for 3 min. The above mixture was added dropwise to a flask under nitrogen protection and reacted for 4 h. After that, 0.5 g of BPO was dissolved in another mixture of 28 g of xylene and 12 g of propylene glycol methyl ether and added dropwise to the flask within 30 min, and the reaction was continued for 90 min, resulting in a colorless and transparent polymer of isobornyl acrylate.

2.2.2 Synthesis of acrylate indole polymers (AIP)

As shown in Scheme 1b, the preparation was carried out by a two-step synthesis. In the first step, 35 g of propylene glycol monomethyl ether, 70 g of xylene, and 5 g of ethyl acrylate were added to 500 mL of a four-necked flask at 95 °C, and then, 5 g of methyl methacrylate, 47.5 g of ethyl acrylate, 26.15 g of glycerol methacrylate, 20 g of xylene, 1 g of 2,4-diphenyl-4-methyl-1-pentene, 1 g of azodiisobutyronitrile, and 4 g of azodiisopentanenitrile were added. After the dropwise addition, weigh 0.5 g of benzoyl peroxide and 10 g of xylene and drop the mixture into a four-necked flask at a uniform rate, and after the dropwise addition is completed,

the reaction is continued for 1.5 h, and the whole process should be carried out under nitrogen. In step 2, 26.15 g of indoleacetic acid (IAA), 37.5 g of xylene, and 25 g of tetrahydrofuran were added to a 95 °C flask, and the reaction was continued for 10 h. The reaction was carried out under nitrogen.

2.2.3 Synthesis of indole isobornyl co-modified acrylic polymers (IAA-IBOMA)

As shown in Scheme 1c, a two-step synthesis was carried out by first adding 28.0 g of xylene with 12.0 g of propylene glycol methyl ether to a four-necked flask at 95 °C, followed by weighing 47.0 g of methyl methacrylate (MMA), 23.0 g of ethyl acrylate (EA), 26.15 g of glycerol methacrylate, 30.0 g of indolyl isobornyl methacrylate (IBOMA), 1 drop of chain transfer agent (MSD), and 1.5 g of azobisisobutyronitrile (AIBN) were mixed; 28.0 g of xylene was added with 12.0 g of propylene glycol methyl ether, and the reaction was stirred for 3 min for 4 h. A nitrogen atmosphere was maintained in the flask. Then, 0.5 g of BPO was weighed

and dissolved in a mixture of 28.0 g of xylene and 12.0 g of propylene glycol methyl ether, and the reaction was continued for 90 min, and finally, 26.15 g of indole acetic acid (IAA), 37.5 g of xylene, and 25.0 g of tetrahydrofuran were added to a three-necked flask at 95 °C, and the reaction was continued for 10 h. Eventually, the solution turned into a dark brown transparent solution.

3 Results and discussion

3.1 Structural characterization and analysis of antifouling coatings with natural extracts

Fourier transform infrared (FT-IR) spectra are shown in Fig. 1a. First, for IBAP, the absorption peaks at 2987, 2950, and 2879 cm^{-1} are due to the stretching vibration of methyl and methylene; the absorption peak at 1725 cm^{-1} is due to the stretching vibration of the ester carbonyl group $\text{C}=\text{O}$ [31]; the absorption peak at 1476 cm^{-1} is due to the deformation vibration of methylene; and the absorption at 1449 and 1388 cm^{-1} peaks is due to the antisymmetric and symmetric deformation vibrations of the methyl group; the absorption peak at 1237 cm^{-1} is from the stretching vibration of C-O linked to the carbonyl group; the absorption peaks at 1177 and 1142 cm^{-1} are caused by the stretching vibration of C-O linked to the alkyl group; the absorption peaks at 1050 and 987 cm^{-1} are due to the stretching vibration of C-C [32], and the absorption peak at 845 cm^{-1} is due to the wobbling vibration of the methyl group; the absorption peak at 750 cm^{-1} is due to the in-plane rocking vibration of methyl group in ethyl group indicating synthesis.

Secondly, the absorption peak of AIP at 3390 cm^{-1} is due to the stretching vibration of hydroxyl; the absorption peak at 3060 cm^{-1} is caused by the stretching vibration of unsaturated C-H; the absorption peaks at 2979 and 2935 cm^{-1}

represent the stretching vibration of methyl and methylene; the absorption peak at 1723 cm^{-1} is due to the stretching vibration of ester $\text{C}=\text{O}$; the absorption peak at 1621 cm^{-1} is from the $\text{C}=\text{C}$ stretching vibration; the absorption peaks at 1447, 1381, and 1342 cm^{-1} are caused by the antisymmetric and symmetric deformation vibration of the methyl group; the absorption peak at 1227 cm^{-1} is due to the stretching vibration of C-O attached to the carbonyl group; the absorption peak at 1156 cm^{-1} is due to the stretching vibration of C-O attached to the alkyl group; the absorption peak at 1018 cm^{-1} represents the stretching vibration of C-C; and the absorption peak at 851 cm^{-1} is due to the stretching vibration of the methyl group. The absorption peak at 851 cm^{-1} is a rocking vibration of the methyl group, and the absorption peak at 743 cm^{-1} is due to an in-plane rocking vibration of the ethyl sub-methyl group and an out-of-plane bending vibration of indole CH indicating AIP synthesis. Finally, it is evident that the IAA-IBOMA curve is not only essentially the same as that of indoleacetic acid but also similar to that of isobornyl, with a few more absorption peaks at 3000–2800 cm^{-1} as compared to indoleacetic acid, which is due to the stretching vibration of saturated alkyl C-H in isobornyl, and at 1050 and 987 cm^{-1} as the stretching vibration of C-C in isobornyl [33, 34]. The absorption peaks in indole acetic acid appeared at 3389, 3055, and 1342 cm^{-1} overlapping many absorption peaks in isobornyl due to the similarity. As shown in Fig. 1, the telescopic vibration absorption peaks of the ester $\text{C}=\text{O}$ and C-O are located at 1724 and 1145 cm^{-1} between the two spectral absorption peaks indicating the esters of the two substances of the overlapping absorption peaks generated the absorption peaks [35]. It means the success of the synthesis.

As shown in Fig. 1b, ^1H NMR spectrum of IBAP shows that the peak at h ($\delta=0.85$ ppm) is the characteristic peak of methyl protons on the aliphatic chain, the peak at j ($\delta=0.90$ ppm) is the characteristic peak of methyl protons

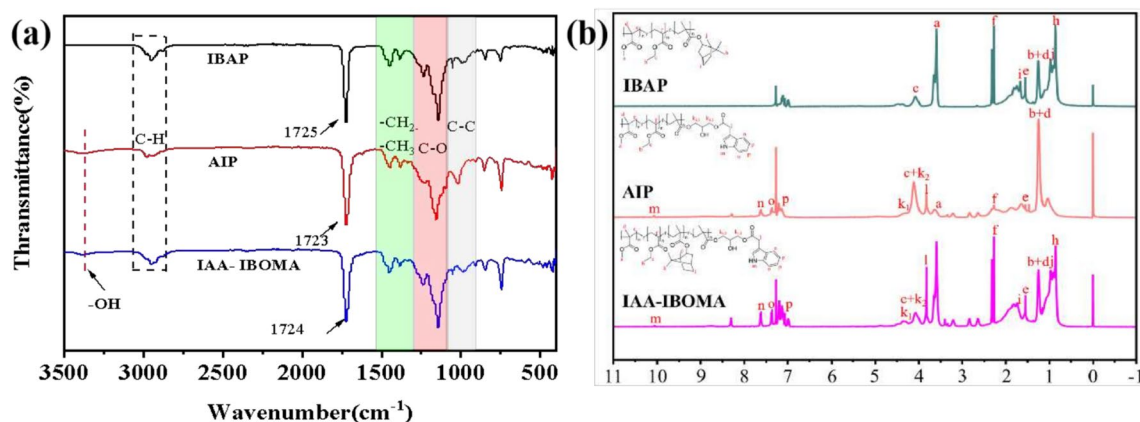


Fig. 1 FT-IR of natural extract-based antifouling coatings (a) and ^1H NMR of natural extract-based antifouling coatings (b)

on the aliphatic chain, and the peak at i ($\delta = 1.66$ ppm) is the characteristic peak of methylene protons on the six-membered aliphatic ring [36]. The above characteristic peaks proved the successful synthesis of IBAP. From the ^1H NMR spectrum of AIP, it can also be observed that the peak at m ($\delta = 10.06$ ppm) is the characteristic peak of amino proton; the peaks at n, o, and p ($\delta = 7\text{--}7.5$ ppm) are the characteristic peaks of aromatic proton; the peak at k ($\delta = 4.4$ ppm) is the characteristic peak of methylene proton connecting carboxyl oxygen; and the peak at l ($\delta = 3.81$ ppm) is the characteristic peak of methylene proton connecting ester carbonyl group with olefin. The peak at l ($\delta = 3.81$ ppm) is the characteristic peak of the methylene proton connecting the ester carbonyl to the olefin. The above characteristic peaks proved the successful synthesis of AIP. The ^1H NMR spectrum of IAA-IBOMA contains both the methyl proton peaks (j and h) and methylene proton peak (i), which belong to IBAP only [37], and the methylene proton peaks (k and l) and aromatic proton peaks (n, o, and p), which belong to AIP only.

3.2 Determination of the morphology of antifouling coatings with natural extracts

The availability of stabilized surface morphology of IBAP, AIP, and IAA-IBOMA antifouling coatings has a

significant impact on their resistance to bioadhesion [38, 39]. The surface morphology of the coatings was characterized by scanning electron microscopy (SEM). As shown in Fig. 2a–c, the AIP, IBAP, and IAA-IBOMA coatings have a stable and smooth morphology. The results of the experimental tests showed that the IBAP, AIP, and IAA-IBOMA coatings possessed stable and smooth surfaces. In addition, to further observe the microstructural dimensions of the natural extract-based materials, the roughness of the material surfaces was evaluated using AFM [40], and the roughness was expressed in terms of R_a (mean value with respect to the reference surface) and R_q (root-mean-square value with respect to the reference surface). As shown in Fig. 2d–f, the R_a and R_q values of IBAP, AIP, and IAA-IBOMA coatings did not vary significantly. The R_a and R_q of the IBAP coating were 0.288 and 0.595 nm, respectively. The R_a and R_q of the AIP coating were 15.1 and 12.2 nm, respectively, while those of the IAA-IBOMA coating were 2.64 and 1.49 nm, respectively. The relatively small changes in these values indicate tighter polymerization after modification. In summary, the test results from SEM and AFM are in good agreement.

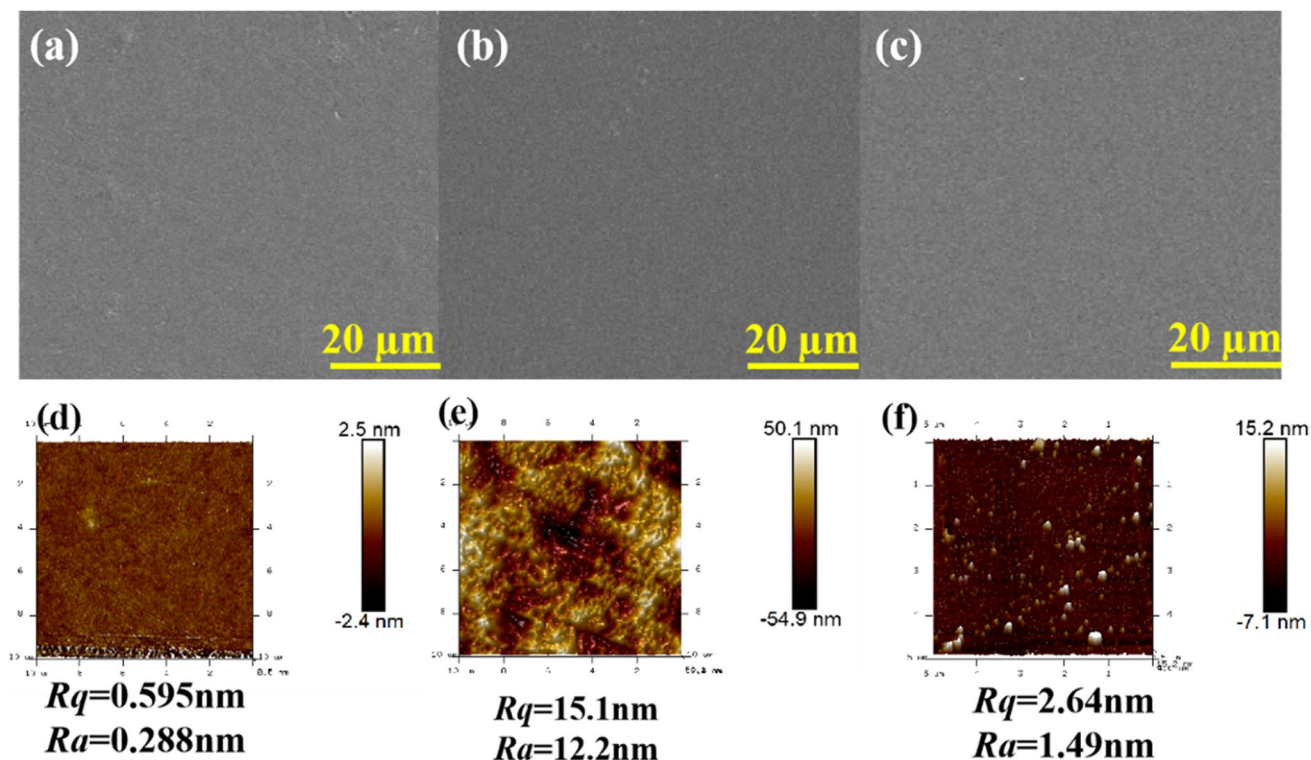


Fig. 2 SEM images of natural extract-based antifouling coatings: **a** IBAP, **b** AIP, and **c** IAA-IBOMA; AFM images of natural extract-based antifouling coatings: **d** IBAP, **e** AIP, and **f** IAA-IBOMA

3.3 Surface wettability test of natural extract-based antifouling coatings

In order to explore the surface wettability properties of natural extract-based antifouling coatings, contact angle (CA) tests were conducted and CA analysis was observed before and after 14 days of immersion in seawater [41–43].

As shown in Fig. 3, the natural extract IBAP exhibits a hydrophobic surface both before and after seawater immersion, while the CA of AIP and IAA-IBOMA remains essentially unchanged at 14-day seawater immersion [44]. This indicates that the IBAP, AIP, and IAA-IBOMA coatings are stable in seawater [45].

3.4 Evaluation of antibacterial properties of natural extract base

To further evaluate the natural extract base for its broad-spectrum antimicrobial and antifouling performance, typical

Escherichia coli and *Staphylococcus aureus* were selected for 24-h culture [46, 47]. The marine bacterium *Pseudoalteromonas* sea was selected to observe the antifouling effect after 48-h incubation [48]. As shown in Fig. 4a, the amount of bacteria on the solid medium in the IAA-IBOMA antifouling coating was greatly reduced, and the bacterial inhibition effect was obvious. As shown in Fig. 4b, the inhibition rates of IBAP, AIP, and IAA-IBOMA antifouling coatings against *Pseudoalteromonas oceanica* were 73.0%, 88.7%, and 95.7%, respectively. Because the IAA-IBOMA component can effectively inhibit bacterial adhesion, it can be seen that the co-modified IAA-IBOM inhibits bacterial adhesion significantly.

3.5 Evaluation of natural extract-based algal inhibition performance

In addition, small crescentic rhodophytes and triangular brown finger algae were used as typical benthic diatom

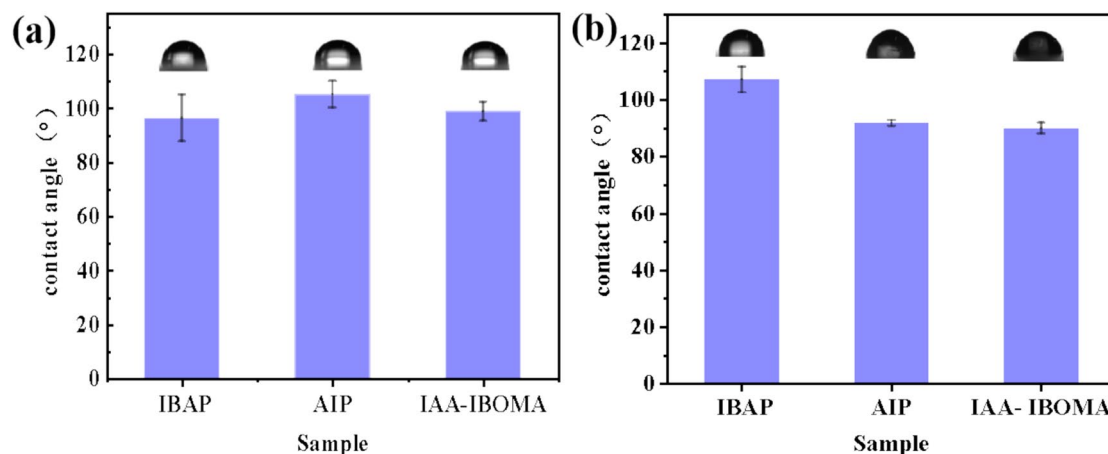


Fig. 3 Water contact angle diagram of natural extract-based antifouling coatings: **(a)** static contact angle and **(b)** contact angle test after 14 days of seawater immersion

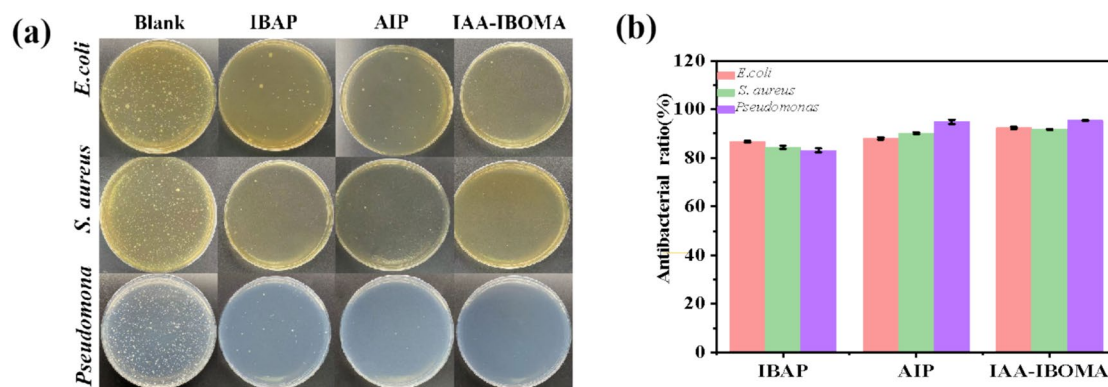


Fig. 4 **(a)** Evaluation of bacterial inhibition performance of natural extract-based antifouling coatings and **(b)** bacterial inhibition rate of natural extract-based antifouling coatings

models to evaluate the antifouling coating's ability to resist diatom adhesion [49, 50]. As shown in Fig. 5a, the number of IBAP, AIP, and IAA-IBOMA adhered in diatoms for 7 days, and the number of IAA-IBOMA diatom adherence was greatly reduced compared with IBAP and AIP. As shown in Fig. 6a, after 7 days of incubation, the algal inhibition rate of IBAP, AIP, and IAA-IBOMA coatings reached 78.06%, 83.54%, and 88.34%, respectively.

For a more detailed examination, fluorescence microscopy was employed to capture images of the IBAP, AIP, and

IAA-IBOMA coatings following a 14-day incubation period. As shown in Fig. 5b, the amount of adhesion on the surface of IAA-IBOMA material was significantly reduced as seen in the digital photograph. As shown in Fig. 6b, the anti-adhesion rates of IBAP, AIP, and IAA-IBOMA coatings reached 80.33%, 86.44%, and 92.00%, respectively. The results proved that the natural extracts showed inhibitory algal effects in both small crescent rhodophytes and triangular brown finger algae sap after 14 days, respectively, and especially the IAA-IBOMA coating had a better effect in preventing bioadhesion [51, 52].

Fig. 5 Fluorescence microscopy of adhesion of natural extract-based antifouling coatings in Rhodophyta minor crescentica and Rhodophyta delta brown finger algae: **a** 7 days and **b** 14 days

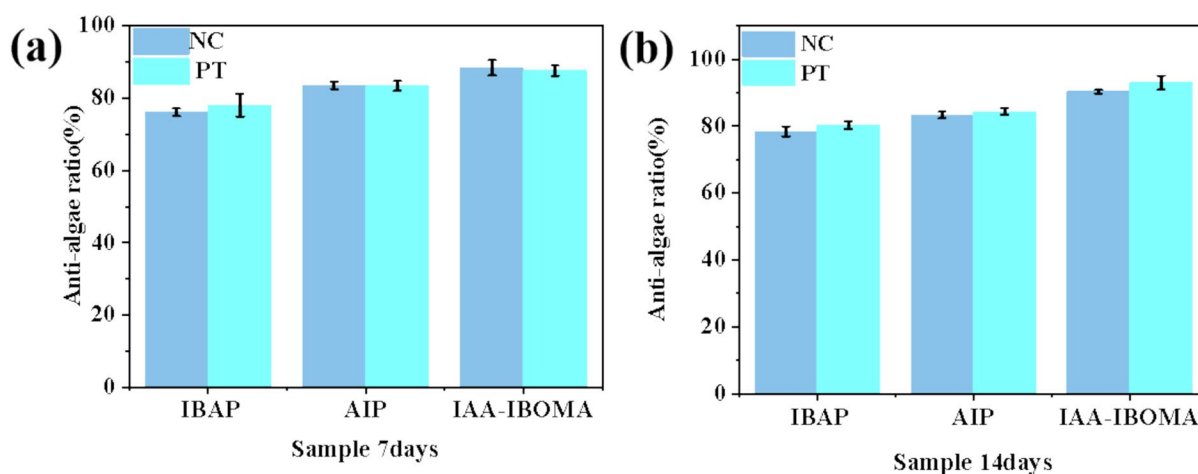
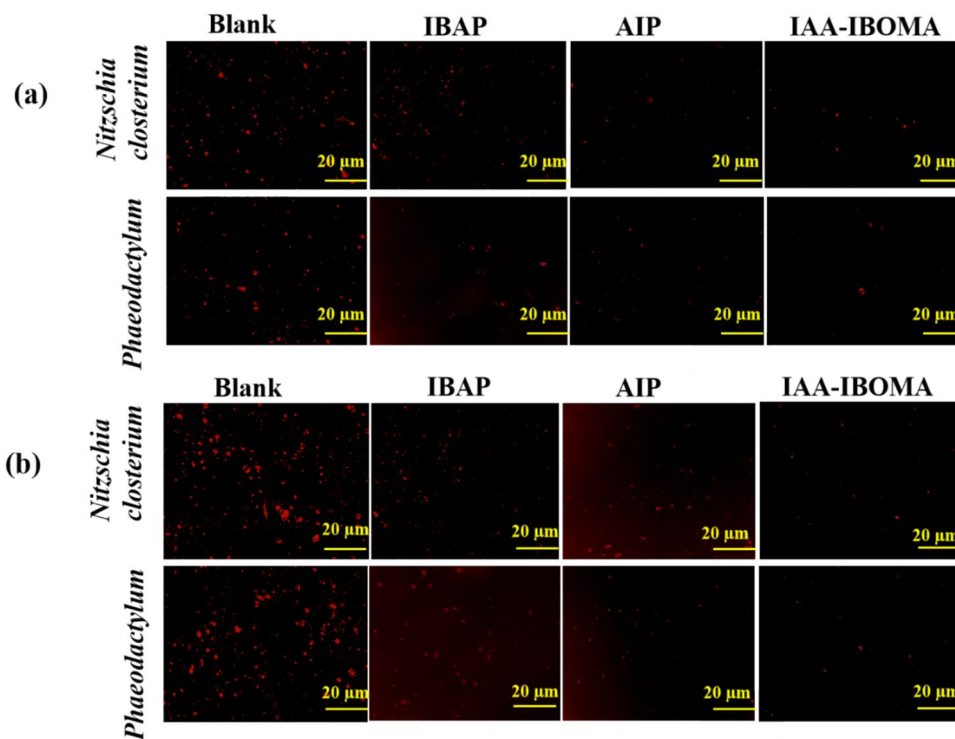


Fig. 6 Inhibition rate of natural extract-based antifouling coatings in Rhodophyta minor crescentica and Rhodophyta delta brown finger algae: **a** 7 days and **(b)** 14 days

3.6 Tests of natural extracts against selective adhesion of mussels

Large biofouling, especially hard-shelled organisms such as mussels, is extremely harmful to ship navigation [53, 54]. Therefore, in this work, we selected the emerald mussels captured at the dock of Haikou to conduct the mussel selective adhesion test to evaluate the antifouling performance, and the mussels were placed on the epoxy resin plate with randomly arranged samples to observe their movement pattern for 24 h, as shown in Fig. 7a; the different composites were randomly arranged in the form of mosaics [55]. The emerald has been cut off the secreted foot filaments, in an unfixed state; at this time, the emerald mussel will secrete the outstretched foot on the surface of the material to test and choose the appropriate surface for attachment. After 24 h, as shown in Fig. 7b–g, after 24 h, emerald mussels 1, 2, and 3 obviously moved their positions and secreted foot filaments to attach to the surface of the material, which coincided with the behavior of emerald mussels that immobilized themselves on the surface of the marine facilities in the marine environment to form marine fouling.

As shown in Fig. 7h, the mussels' attachment on blank glass slides was significantly higher than the number of attached pedofilament disks on the surfaces of the IBAP,

AIP, and IAA-IBOMA antifouling coatings [56, 57]. This demonstrates that emerald mussels selectively chose the more “suitable” surface of the glass slides and avoided the surfaces of the IBAP, AIP, and IAA-IBOMA materials. In particular, IAA-IBOMA had a significant repellent effect. The results showed that IAA-IBOMA had excellent performance in inhibiting or avoiding mussel adhesion to its surface [58, 59].

3.7 Environmental toxicology tests

To further validate the environmental toxicology test, environmental toxicology studies were carried out on IBAP, AIP, and IAA-IBOMA, as shown in Fig. 8a; the absorbance of the algae night was incubated with IBOMA superimposed with varying number of coatings (1–8) pieces for 2–14 days, and the diatoms grew exponentially from the 6th day onwards [60–62]. It can be seen that the more the number of coatings superimposed, the smaller the number of OD values in diatoms [63]. As shown in Fig. 8b, the inhibition effect of IAA-IBOMA coating against target organisms was obvious, and the results verified that IAA-IBOMA had excellent antifouling ability [64–66].

As shown in Fig. 9, the color of diatom growth is consistent with the above OD results. It was clearly seen that

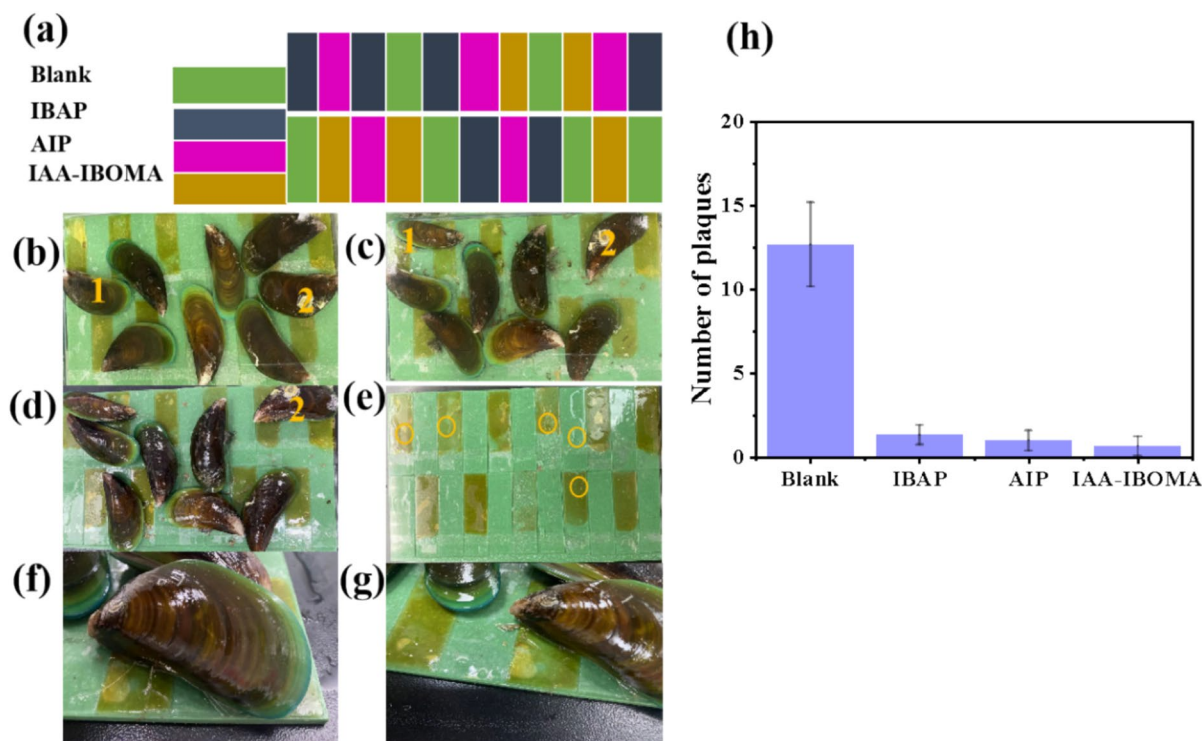


Fig. 7 Selective adhesion test results of emerald mussel: **a** schematic diagram of mosaic plate arrangement (**b–g**) 0 h, 12 h, and 24 h in the process of emerald mussel adhesion and movement on the surface

of the natural extract-based antifouling coating; **h** statistical diagram of emerald mussel footsilk disk on the surface of the natural extract-based antifouling coating

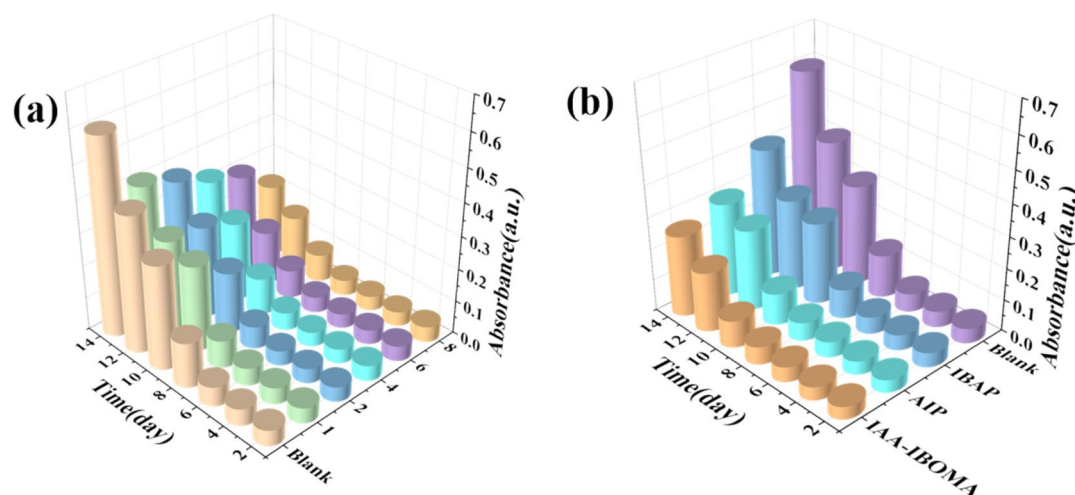


Fig. 8 **a** Absorbance of different amounts of IBAP in *Rhodobacter crescentus*; **b** 14-day growth curves in *Rhodobacter crescentus* sap with natural extract-based antifouling coating

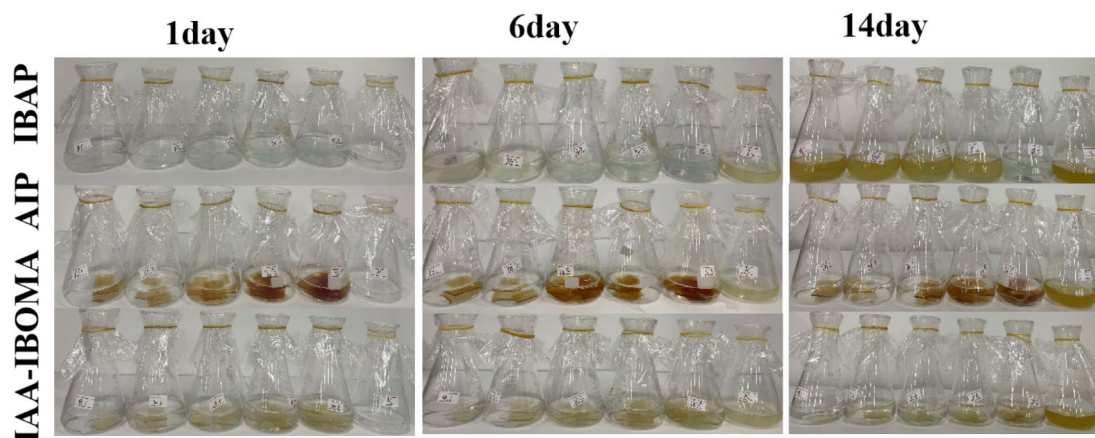


Fig. 9 Color change of algal liquid of *Rhodophyta crescentica* with different quantities of natural extract-based antifouling coatings at 1 day, 7 days, and 14 days

1, 2, 4, and 6 tablets of IBAP grew the algae to the original yellow color in the same number of days of incubation, while IBAP with 8 tablets obviously inhibited the growth of diatoms resulting in colorless. This indicates that the natural extracts have a strong inhibitory effect on diatoms, while the OD value of the test showed that the concentration was very low and the amount of algae was consistent with that observed by the naked eye. The OD values of AIP and IAA-IBOMA tests, together with the picture results, verified that the modified polymer antifouling coatings had a more prominent inhibitory effect on diatoms [67–69]. It shows that IBAP, AIP, and IAA-IBOMA have an inhibitory effect on small crescent-shaped algae on the surface of contact coating and hardly inhibit the growth of small crescent-shaped algae in the environment.

4 Conclusion

In summary, three polymer coatings based on acrylic acid grafted with indole and isobornyl-modified polymers, namely isobornyl acrylate polymer (IBAP), indole acetic acid modified polymer of acrylic acid (AIP), and indole isobornyl co-modified polymer of acrylic acid (IAA-IBOMA) coatings, have been prepared by free radical polymerization method [70–73]. The SEM and AFM characterization illustrated that the IBAP, AIP, and IAA-IBOMA materials had good film-forming, smooth, and flat surfaces. The anti-adhesion properties of indole acetic acid and ice tablet base were investigated by bacterial and algal inhibition experiments. The modification of AIP and

IAA-IBOMA materials greatly reduced the adhesion of bacteria and diatoms, as follows: the bacterial inhibition rate of IAA-IBOMA reached 95.7%. The inhibition rate of the adhesion of small crescentic lozenge algae and triangular brown finger algae reached 92%. The anti-mussel selective adhesion inhibition rate reached 93.3%, showing outstanding anti-bio-adhesion performance. In addition, the IAA-IBOMA coating showed excellent environmental friendliness and long-term stability, making it suitable for marine environments [74–76]. Ultimately, this work will contribute to the promising application of natural extract antifouling coatings in the marine environment.

Author contribution Conceptualization: P.Z. and L.Z.; methodology: M.L., W.M., P.Z., and L.Z.; validation: M.Z. and S.L.; formal analysis: M.L., S.H., and J.L.; investigation: J.L. and S.L.; writing—original draft preparation: M.L., W.M., and P.Z.; writing—review and editing: P.Z. and L.Z.; supervision: P.Z. and L.Z.; project administration: P.Z. and L.Z.; funding acquisition: W.M., S.H., and L.Z. All authors have read and agreed to the published version of the manuscript.

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Data availability Data will be made available on request.

Declarations

Competing interests The authors declare no competing interests.

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